MIL-T-22012 (NOrd)

8 June 1959

MILITARY SPECIFICATION

TIMER, SEQUENTIAL, BATTERY CONDITIONING, TD-267/U

1. SCOPE.

1.1 This specification covers the requirements for the production, inspection, and test of a sequential timer to be provided in TALOS shipboard installations for charging, discharging, shorting, and recharging, the nickel-cadmium type batteries contained in TALOS Guided Missiles. This specification describes equipment to be used aboard the CLG-4, CLG-5, and other Naval vessels, and at Naval depots and training schools.

2. APPLICABLE DOCUMENTS.

2.1 The following specifications, standards, publications and drawings of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-W-5086	Wire, Electrical 600-Volt, Copper, Aircraft

FSC 4935

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:53Z
Check the source to verify that this is the current version before use.

MIL-T-22012 (NOrd)

MIL-R-15109	Resistors and Rheostats, Shock-proof (Naval Shipboard Use)
MIL-S-15291	Switches, Rotary, Snap Action
MIL-T-17296 (NOrd)	Test and Checkout Equipment, for Use with Guided Missiles (Fixed Installation): General Specification
MIL-C-17588	Circuit Breaker (Automatic-ALB) and Switch, Toggle (Circuit Breaker, Non-Automatic NLB) Air, Insulated Enclosure, 125 Volts and Below AC or DC, Naval Shipboard
MIL-R-18546	Resistor, Fixed, Wire Wound, Power Type, Chassis Mounted

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	General Instructions for the Design, Manufacture, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:53Z
Check the source to verify that this is the current version before use.

requirements specified on the drawings shall ... ance with the detailed ...

	Inspection of Naval Ordnance Equipment
OP 2230	Design Practices for Electronic Equipment
OP 2756	Timer, Sequential, Battery Conditioning TD-267/U
NAVORD OS 6341	Miscellaneous General Ordnance Design Requirements
NAVORDINST 4100.4	Critical Materials; Policy for Use in Design and Construction of Naval Ordnance and Action Required
NAVORDINST 4121.6	Specifications: Guide for the Preparation of Section 5, "Preparation for Delivery"
NAVORDINST 5600.14	Technical Documentation

DRAWINGS

BUREAU OF ORDNANCE

LD 492341	Timer, Sequential, Battery Conditioning TD-267/U

(Copies of specifications, standards, publications and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 202640-19T10:53Z
Check the source to verify that this is the current version before use.

2.2 Requests for deviation. - Requests for deviation from this specification, applicable specifications, and publications specified herein shall be as set forth in NAVORD OP 400.

2.3 Precedence. - When the requirements of the contract or order, this specification, or subsidiary specifications are in conflict, the procurement document of higher precedence shall govern. Precedence of documents shall be in accordance with the paragraph entitled "Order of Precedence" in NAVORD OP 400.

3. REQUIREMENTS.

3.1 General description. - The sequential timer provides the necessary components and associated circuitry for automatic and manual control of charging or discharging, shorting, or recharging of the main power batteries and telemetering batteries used in TALOS missiles. The sequential timer will be located in the ship's battery room and will be used in conjunction with the PP-1458/U power supplies. A timing mechanism, of electronic or electro-mechanical design, is included in the unit to provide automatic control of the following three-step cycle: (1) a discharging period variable between 5 and 60 minutes, (2) a shorting period of approximately one hour, and (3) a charging period of approximately one hour. Battery discharge is accomplished by means of load resistors which control the discharge rate. During the discharge cycle, when the load voltage has reached a predetermined level, the timer mechanism automatically switches over to the shorting circuit. At the completion of the short circuiting period, the timer mechanism causes the energized power supply source to be automatically switched into the circuit to recharge the batteries. The necessary adaptors, and terminal boards are provided to facilitate connections from the 9.3 and 35.65 v.d.c. rectifier power sources to the missile battery assemblies for charging. Means are provided for manual or automatic operation of the timer unit. In manual operation the operator can select charge, discharge, or short circuit funtions for the 6.6 v. or 26.5 v. battery assemblies. In automatic operation the timer mechanism is activated so that the charge, discharge, or short circuit

Source: https://assist.dla.mil -- Downloaded: 202640-19T10:53Z
Check the source to verify that this is the currentversion before use.

functions are performed in automatic sequence. Circuit breakers or reverse current and anti-short interlocks in the timer prevent damage to the battery due to inadvertent switching from one function to the other. Two voltmeters in the unit enable continuous monitoring of battery voltages during the charge cycle. Means are provided to enable the selection of load resistors. A detailed description of the timer is contained in OP 2756.

3.2 General requirements.

3.2.1 Procedures. - For applicable requirements not covered herein or in the applicable drawings and specifications listed on BuOrd LD 492341 the requirements of MIL-T-17296 (NOrd) shall apply.

3.2.2 Design and construction. - The sequential timer shall be enclosed by an aluminum cabinet and at the option of the procuring activity shall be adaptable to either bench or bulkhead mounting. Parts shall be readily replaceable without causing damage to adjacent parts or wiring. It shall be the contractor's responsibility to provide connections compatible with TALOS shipboard cabling provisions for the power supply rectifiers. The sequential timer shall be capable of cycling the 6.6 v. and 26.5 v. batteries simultaneously and shall be capable of being used with TALOS Mk 11 Mod 2 and Mod 4 missile battery packs. The unit shall operate satisfactorily on 115 v., 60 cps power. Insofar as is commensurate with the state of the art the design shall provide for:

(a) simple packaging
(b) accessibility for servicing
(c) rugged parts
(d) consistent and appropriate tolerances
(e) adequate heat dissipation
(f) simple production and production test procedures
(g) minimum use of critical materials

3.2.3 Resistors and rheostats. - All resistors and rheostats

Source: https://assist.dla.mil -- Downloaded: 202640-19T10:53Z
Check the source to verify that this is the current version before use.

used in the battery discharge circuitry shall meet or exceed the requirements of Specification MIL-R-15109, or Specification MIL-R-18546.

3.2.4 Materials. - Unless otherwise specified in the contract, requisition or purchase order, all materials and parts entering into the final assembly of the sequential timer shall conform strictly to the detailed requirements indicated on the drawings listed on BuOrd LD 492341.

3.2.4.1 Critical materials. - The attention of the contractor is directed to the responsibilities and reports which may be invoked by the Bureau of Ordnance or its designated representative as set forth in NAVORDINST 4100.4 with regard to critical materials.

3.2.5 Finishes, coatings, construction and dimensions. - Finishes, coatings, construction and dimensions for the sequential timer shall be strictly in accordance with the applicable drawings listed on BuOrd LD 492341.

3.2.6 Moisture, fungus and corrosion. - The sequential timer shall be rendered moisture, fungus and corrosion resistant in accordance with the detailed requirements specified on the drawings listed on BuOrd LD 492341. When moisutre, fungus, or corrosion resistance is not specified in these drawings, the timer shall be moisture, fungus and corrosion resistant as required by NAVORD OS 6341.

3.2.7 Interchangeability. - Like assemblies, sub-assemblies, and replaceable parts shall be electrically, mechanically and functionally interchangeable regardless of the manufacturer or supplier. They shall be easily interchangeable without physical or electrical modifications.

3.2.8 Reliability. - It is imperative that reliability of operation be considered of prime importance in the fabrication and assembly of

Source: https://assist.dla.mil -- Downloaded: 20261O-19T10:53Z
Check the source to verify that this is the current version before use.

the sequential timer.

3.2.9 Marking. - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed on BuOrd LD 492341.

3.3 Detail requirements.

3.3.1 Components. - The sequential timer shall include the following components:

(a) Timer mechanism
(b) Load resistors
(c) Circuit breakers (Optional)
(d) "AN" connectors
(e) Adaptor cables
(f) Terminal boards
(g) Voltmeters
(h) Switching devices

3.3.2 Timer mechanism. - The timer mechanism shall be of either electronic or electro-mechanical design at the option of the procuring activity. The timer shall provide automatic programming of the recharge cycle described in 3.1, and shall provide automatic shut-off of power to the batteries at the completion of each charging cycle. In addition, the unit shall contain provisions for adjustment of the time duration of each individual cycle. The timing mechanism shall be capable of operating on 115 v., 60 cps power.

3.3.3 Load resistors. - The discharge circuits of the timer shall include load resistors which simulate counterpart missile loads. In lieu of resistors, the contractor may substitute two rheostats variable over the range of the 6.6 v. and 26.5 v.d.c. loads of TALOS Guided Missiles.

3.3.4 Circuit breakers. - If circuit breakers are provided in

Source: https://assist.dla.mil -- Downloaded: 20261O-19T10:53Z
Check the source to verify that this is the current version before use.

the circuits of the sequential timer, the operating mechanism shall be such that the rupturing contacts shall be quick-make and quick-break under all conditions of manual and automatic operation. The circuit breakers shall provide protection against damage to components, as well as against injury to personnel, under conditions of overload or inadvertent switching to an out-of-cycle function during manual operation. All circuit breakers shall conform to the requirements for Type ALB-10 circuit breakers as set forth in Specification MIL-C-17588.

3.3.5 "AN" connectors. - "AN type connector plugs in the lower portion of the front of the cabinet shall provide for the connection of adaptor cables extending from the battery timer to the missile battery assemblies. It shall be the contractor's responsibility to insure that the "AN" type connectors of the adaptor cables are compatible with those of the missile battery assemblies.

3.3.6 Adaptor cables. - Three (3) adaptor cables, with "AN" connectors permanently affixed, shall be provided for connection of the sequential timer to the missile battery assemblies. Each adaptor cable shall be two and one-half feet in length. All adaptor cables shall be flexible, heat and oil resistant, be provided with a sheath which is impervious to water, and conform to the requirements of Specification MIL-W-5086. The cables shall have a minimum rating of 75 amperes and 600 volts at an ambient temperature of 40°C. It shall be the contractor's responsibility to insure that the adaptor cable conductors are compatible with those provided on the missile battery assemblies.

3.3.7 Terminal boards. - Terminal boards shall be provided in the sequential timer to facilitate connection to the shipboard power sources (d.c. rectifiers PP/1458 U Type I and Type II). In addition, a terminal board shall be provided to enable external sensing of the 9.3 and 35.65 v.d.c. inputs.

3.3.8 Voltmeters. - Two ruggedized plus or minus one (1) percent tolerance voltmeters shall be provided to enable a visual indication of the battery voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:53Z
Check the source to verify that this is the current version before use.

3.3.9 Switching devices. - Switching devices and relays shall be provided in the sequential timer to enable switching from one function to another.

3.4 Technical documentation. - Documentation of the equipment described herein shall be as specified in the contract, requisition or purchase order. All documentation required by said contract or order shall be prepared in accordance with the basic guides and schedules set forth in NAVORDINST 5600.14.

3.5 Cleanliness. - The standard of cleanliness throughout the equipment shall be commensurable with good manufacturing practices. There shall be no foreign matter such as excess solder, bits of insulation, filings, chips, etc., left within the assembly.

3.6 Workmanship. - Workmanship shall be first class in every respect. Refer to NAVORD OP 2230 for design and workmanship practices for electronic equipment. The requirement for first class workmanship does not preclude the use of time and labor-saving techniques commensurate with the nature of the operation of the sequential timer, provided reliability is in no way jeopardized. A sample of each timer may be selected jointly by the Government Inspector and the contractor to serve as a standard of acceptable workmanship.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS.

4.1 General. - Unless otherwise specified, the sequential timer shall be inspected and tested in accordance with the requirements of this specification and the applicable requirements of MIL-T-17296 (NOrd). Satisfactory tests, including environmental tests, and tests to determine suitability of electrical and mechanical characteristics, and satisfactory test data which indicate compliance with the specified requirements and suitability for the intended application are required for acceptance of the sequential timer or any of its components. Failure to meet the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:53Z
Check the source to verify that this is the current version before use.

specified requirements and tests shall require the contractor to effect the necessary adjustments, modifications, repairs, or replacements. Incompatibility with any of the requirements of this specification shall be considered cause for rejection.

4.2 Contractor's inspection. - NAVORD OP 400 is applicable. Inspection records of the examinations and tests shall be kept complete and available to the Government when specified in the contract or order.

4.3 Government inspection. - The Government Inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in accordance with the instructions contained in NAVORD OP 400. Government inspection will include the requirements of the following paragraphs of this specification:

a.	3.2.1	Procedure
b.	3.2.4	Materials
c.	3.2.5	Finishes, Coatings, Construction and Dimensions
d.	3.2.6	Moisture, fungus and corrosion
e.	3.2.9	Marking
f.	3.3.5	AN connectors
g.	3.3.6	Adaptor cables
h.	3.3.7	Terminal boards
i.	3.3.8	Voltmeters
j.	3.3.9	Switching devices
k.	3.3.4	Technical documentation
l.	3.5	Cleanliness
m.	3.6	Workmanship

4.4 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection shall be in accordance with the provisions set forth in MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:53Z
Check the source to verify that this is the currentversion before use.

4.5 Classification of tests. - Unless otherwise specified by the procuring activity three types of tests shall be performed - design evaluation tests, acceptance tests, and quality assurance tests. General descriptions of these tests are given in 4.5.1, 4.5.2, and 4.5.3 respectively. When applicable, each unit shall perform satisfactorily before, during, and after each test. Satisfactory performance shall be that performance which conforms to the requirements of section 3 or that which conforms to the acceptance standards approved by the procuring activity. These tests normally will be performed at the place of manufacture by or under the supervision of an agent of the procuring activity, or at an Armed Services installation at the discretion of the procuring activity.

4.5.1 Design evaluation tests. - Design evaluation tests are tests performed on one or more timers to verify its ability to meet the design and performance requirements of this specification. Test procedures shall insure the adequacy of the timer to comply with the requirements of section 3. All of the Design Evaluation Tests shall be performed on the same timer. A design evaluation test procedure shall be prepared by the contractor and submitted to the procuring activity for approval at least 60 days prior to submission of the equipment for design evaluation.

4.5.1.1 Unless otherwise specified, the following design evaluation tests shall be performed on the set:

(a) temperature tests
(b) humidity tests
(c) inclination test
(d) shock tests
(e) vibration tests
(f) enclosure test (protected areas)
(g) electrical interference tests
(h) mechanical noise test
(i) accelerated life tests

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:53Z
Check the source to verify that this is the currentversion before use.

4.5.2 Acceptance tests. - Acceptance tests are those tests performed on each unit to insure satisfactory performance and the greatest possible reliability of the timer. Each timer must pass the acceptance test before it is accepted by the procuring activity. In general, acceptance tests are less severe than design evaluation tests. The contractor shall prepare an acceptance test procedure and submit it to the procuring activity for approval not less than 60 days prior to submission of the equipment for acceptance.

4.5.2.1 Unless otherwise specified, the following acceptance tests shall be performed on the set:

(a) safety tests (see MIL-T-17296)
(b) control and control circuit tests (see 3.1 and 3.3.2)
(c) power tests (see MIL-T-17296)
(d) variation in line voltage and frequency tests (see MIL-T-17296)
(e) fusing tests (see MIL-T-17296)

4.5.3 Quality assurance tests. - Quality assurance tests are those tests performed on an arbitrary timer selected by an agent of the procuring activity to insure continued quality of the timer unit and to demonstrate consistancy of production techniques. These tests will be performed to the same procedures and acceptance standards as design evaluation tests.

4.5.4 Reports. - Unless otherwise specified by procuring activity, after the completion of each test, the contractor shall submit to the procuring activity four copies of the test results including all details necessary for proper interpretation thereof.

4.6 Test equipment.

4.6.1 Test and inspection equipment and facilities. - The contractor shall furnish and maintain all necessary test equipment and facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and unless definite requirements are specified,

Source: https://assist.dla.mil -- Downloaded: 20261O-19T10:53Z
Check the source to verify that this is the currentversion before use.

they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.6.2 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

4.7 Preservation, packaging, packing and marking.

4.7.1 Inspection and tests. - The contractor shall perform Type II Acceptance Inspection and Tests in accordance with Specification MIL-P-116. Type I Preproduction Inspection and Tests shall be required when specified by the procuring activity.

4.7.2 Group tests. - The following group tests shall be performed by the contractor in accordance with Table III of Specification MIL-P-116.

4.7.2.1 Group A tests. - All visual inspection shall be performed as specified.

4.7.2.2 Group B tests. - Tests for determination of cleanliness, air-water-vacuum-seal leakage, and heat seal shall be performed when specified by the procuring activity.

4.7.2.3 Group C tests.

4.7.2.3.1 Cyclic exposure test. - When specified by the procuring activity cyclic exposure test A shall be performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:53Z
Check the source to verify that this is the current version before use.

4.7.2.3.2 Rough handling test. - When specified by the procuring activity, the free fall drop test, incline impact test, and edgewise drop test shall be performed.

5. PREPARATION FOR DELIVERY.

5.1 General. - The requirements specified herein apply only to direct purchase by or direct shipment to the Government procuring activity. The levels of preservation and packaging and packing are contained in NAVORDINST 4121.6. The selection of applicable levels shall be specified in the contract or order.

5.2 Preservation and packaging.

5.2.1 Level A.

5.2.1.1 Cleaning. - The equipment shall be cleaned in accordance with Method C-1 of Specification MIL-P-116.

5.2.1.2 Drying. - The equipment shall be dried by drying procedures D-1 or D-4 as set forth in Specification MIL-P-116.

5.2.1.3 Preservative application. - Unless otherwise specified by the procuring activity, a preservative shall be applied to the equipment in accordance with Specification MIL-P-116. Selection of the preservative shall be in accordance with Table I of Specification MIL-P-116.

5.2.1.4 Unit packaging. - When a preservative is applied as specified in 5.2.1.3, the equipment shall be packaged in accordance with Method IIb of Specification MIL-P-116. When a preservative is not applied, the equipment shall be packaged in accordance with Method III of Specification MIL-P-116.

5.2.1.4.1 Intermediate packaging. - Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:53Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. - Not applicable.

5.2.3 Level C. - When authorized by the procuring activity, cleaning, drying, preservative application, and packaging shall be in accordance with the manufacturers' commercial practice.

5.3 Packing.

5.3.1 Level A. - Equipment packaged as specified in 5.2.1.4 shall be packed in cleated plywood or nailed wood boxes in conformance to Specification PPP-B-601, Type III Class I, Styles A or B (Overseas Type) or PPP-B-621, Class 2, Type II load, Style 2, respectively.

5.3.2 Level B. - Not applicable.

5.3.3 Level C. - Equipment packaged as specified in 5.2.3, shall be packed to assure protection to prevent damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the modes of transportation.

5.4 Marking.

5.4.1 Normal marking. - In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES.

6.1 Intended use. - The sequential timer is intended for TALOS shipboard use to implement the requirements for periodic recharging of the missile main power batteries, and for recharging telemetering batteries when required.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T10:53Z
Check the source to verify that this is the currentversion before use.

6.2 Warning to contractors. - Where requirements of this specification refer to information to be furnished by the procuring activity on request, it shall be the responsibility of the contractors to request this information from the procuring activity.

6.3 Special contractual requirements. - Special note should be made of MIL-T-17296 (NOrd) wherein items to be covered in the invitation for bids, contracts and orders are listed. These items are in addition to any contained in this specification.

Patent Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:53Z
Check the source to verify that this is the current version before use.